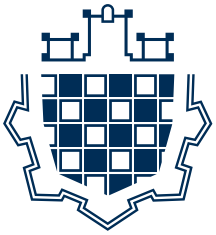

UNIVERSITY OF MANNHEIM

School of Business Informatics and Mathematics

Test Suite Minimisation in LASSO Using Stimulus-Response Matrices

Bachelor Thesis

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Abstract

Large-scale industrial software systems face significant challenges in executing test suites efficiently. In a commercial SAP environment, Bach et al. report test suites comprising more than 130,000 tests requiring hundreds of hours of sequential runtime [2]. Such execution times render continuous testing computationally infeasible and economically prohibitive. The study further revealed extensive redundancy within these test suites, implying that substantial execution cost yields only marginal additional coverage benefit. However, this is not an isolated case, test-suite redundancy is a widespread and well-documented issue in industrial software development. While this computational burden cannot be eliminated entirely, its impact can be substantially reduced through test-suite minimisation, which systematically removes redundant tests with respect to defined adequacy criteria such as statement coverage, branch coverage, or mutation score [20]. However, naive reduction approaches risk compromising fault-detection effectiveness, as tests with unique defect-revealing capabilities may be inadvertently excluded.

This thesis investigates test-suite minimisation in the context of the *Large-Scale Software Observatorium* (LASSO) [12], a research platform for automated selection, execution, and comparative analysis of software systems. LASSO integrates open-source repositories and combines static and dynamic analyses within a unified framework designed for language independence. Its current implementation focuses on Java and Python systems and enables empirically grounded studies of *Big Code* ecosystems (large, diverse corpora of source code enriched with metadata such as bug-fix histories, version control traces, and execution outcomes [1]).

Our approach leverages *Stimulus–Response Matrices* (SRMs) [12], a two-dimensional representation mapping test stimuli to observed execution outcomes. We augment SRM entries with per-test coverage and mutation information, enabling minimisation algorithms to operate on fine-grained behavioural data rather than language-specific code structures.

Through a targeted literature survey encompassing greedy heuristics, exact op-

timisation, search-based, and data-driven minimisation strategies, we comparatively assessed candidate algorithms along seven established criteria: SRM compatibility, coverage retention, reduction effectiveness, fault-detection retention, mutation-coverage preservation, computational scalability, and implementation feasibility. The Mutation-Aware Enhanced HGS (MA-EHGS) algorithm emerged as the most promising candidate. MA-EHGS extends the empirically validated Enhanced HGS baseline [7] by integrating mutation weighting to balance coverage preservation and fault-detection power [9].

We successfully integrated MA-EHGS into LASSO and validated the implementation on real-world SRMs. Our evaluation achieved reductions within the ranges reported in prior studies, while maintaining high coverage and mutation-scores [7, 9]. The algorithm executes in $O(nm)$ time, ensuring computational feasibility for large-scale software analysis workflows.

Contents

List of Figures

List of Tables

1. Introduction

Software testing ensures that program modifications do not introduce defects. As systems evolve, test suites expand, increasing execution cost and maintenance effort. Suites often combine human-written, automatically generated, and, increasingly, AI-generated tests. Tools such as Randoop [16] and EvoSuite [6] can generate thousands of cases, and in large projects, execution may require days or weeks [18]. While comprehensive suites achieve high coverage and mutation scores, they frequently contain redundant tests whose requirements are subsumed by others, causing prolonged execution, higher costs, and elevated maintenance overhead.

Test-suite minimisation systematically removes redundant tests while preserving fault-detection capability [20]. Techniques can be broadly divided into *coverage-preserving* approaches, which guarantee that all coverage requirements remain satisfied, and *coverage-relaxing* approaches, which permit limited coverage loss in exchange for stronger reduction. Minimisation can greatly decrease computational demand and feedback latency, but excessive reduction—especially in coverage-relaxing strategies—risks removing tests with unique defect-revealing potential. Effective minimisation therefore requires a careful balance between reduction efficiency and fault-detection retention.

1.1. Background and Motivation

Testing remains a major cost driver in software engineering. Studies estimate that verification and validation account for up to half of total development effort [3]. Although these figures originate from traditional development models, the cost pressure persists in modern continuous integration, where automated or AI-generated suites may contain tens of thousands of tests [18, 2]. Executing all tests after each modification is often infeasible, motivating scalable approaches

that reduce suite size without compromising quality. Test-suite minimisation addresses this challenge by identifying and removing redundant tests while retaining those essential for coverage and fault detection. As an optimisation problem, it is NP-hard [20] and therefore typically solved through heuristics or approximation algorithms, which are further analysed in Chapter 2.

1.2. LASSO Context

The *Large-Scale Software Observatorium* (LASSO) [12] is a research platform developed by the Chair of Software Engineering at the University of Mannheim. It provides an environment for the scalable analysis and observation of large software corpora, commonly referred to as *Big Code*. LASSO automates the collection and analysis of both static information (e.g., source code properties) and dynamic information (e.g., run-time execution behaviour) from extensive sets of executable open-source systems. Its domain-specific scripting language, the *LASSO Scripting Language* (LSL), enables researchers to define reproducible pipelines for tasks such as code search, automated unit-test generation, and large-scale *Test-Driven Software Experiments* (TDSEs). These capabilities make LASSO a central infrastructure for empirical software engineering and for evaluating emerging technologies, including Generative AI for code.

LASSO employs *Stimulus–Response Matrices* (SRMs) [12] as a unified representation for systematic comparison. An SRM is a two-dimensional matrix in which rows denote test stimuli, columns denote systems, and cells contain structured response objects that capture execution outcomes. From this representation, coverage metrics, mutation scores, and behavioural characteristics can be systematically extracted and analysed, enabling language-independent comparison of functionally equivalent implementations. Section 2.2 provides detailed SRM definitions and explains coverage and mutation data extraction.

While LASSO enables comprehensive behavioural and coverage analysis, it currently lacks an integrated mechanism for test-suite minimisation. This absence can lead to redundant test executions that accumulate significant computational overhead when scaled across the platform's extensive corpus. Incorporating minimisation within LASSO would help mitigate such redundancy while maintaining the behavioural coverage required for reliable system analysis.

1.3. Research Objectives and Questions

This thesis investigates the integration of test-suite minimisation within the LASSO framework. Specifically, it aims to design, evaluate, and implement an SRM-based minimisation approach that maintains language independence and scalability. The following research questions guide the study:

1. **RQ1:** Which test-suite minimisation techniques demonstrate the highest effectiveness and practical applicability according to current literature?
2. **RQ2:** What evaluation criteria are most appropriate for assessing minimisation techniques within LASSO's SRM-based environment?
3. **RQ3:** Which algorithmic approach best balances reduction effectiveness, coverage preservation, and computational scalability for integration with LASSO?
4. **RQ4:** How can the selected approach be integrated into LASSO's analysis pipeline while maintaining language independence?

1.4. Implementation Strategy

This thesis implements minimisation directly on LASSO's SRM abstraction rather than integrating external tools. Existing frameworks (OpenClover, PIT) support limited languages and require source code access or language-specific instrumentation, conflicting with LASSO's language-independent design.

SRMs already encapsulate all information necessary for behavioural minimisation: test identifiers, execution results, coverage metrics, and mutation data. Operating exclusively on this structure maintains language independence, ensures scalability by processing compact SRM representations instead of source artefacts, and promotes architectural cohesion through seamless integration with LASSO's analysis pipeline.

1.5. Thesis Structure

The remainder of this thesis is organised as follows:

Chapter 2 introduces coverage, mutation testing, SRMs, and formally defines the minimisation problem.

Chapter 3 reviews related work across greedy, exact, search-based, and data-driven techniques.

Chapter 4 defines evaluation criteria (C1–C7) and presents the scoring framework for algorithm comparison under SRM constraints.

Chapter 5 applies the framework to rank algorithms and justify the selection of the Mutation-Aware Enhanced HGS approach.

Chapter 6 details the implementation and integration into LASSO.

Chapter 7 discusses limitations, trade-offs, and deployment considerations.

Chapter 8 concludes and outlines future work.

2. Fundamentals

This chapter establishes the theoretical foundation for this work. It introduces the core concepts of test coverage and mutation testing, explains the *Stimulus–Response Matrix* (SRM) representation used within LASSO, and formalises the test-suite minimisation problem, including trade-offs between coverage preservation and reduction strength.

2.1. Test Coverage and Mutation Score

Test coverage quantifies the proportion of program elements exercised by a test suite [21]. Coverage metrics indicate structural thoroughness but provide limited insight into fault-detection capability.

2.1.1. Coverage Metrics

Coverage criteria define program "parts" at different granularity levels [21]. Common coverage types include:

- **Instruction Coverage:** proportion of bytecode instructions executed.
- **Line Coverage:** proportion of source lines executed.
- **Branch Coverage:** proportion of conditional branches whose true/false outcomes are both executed.
- **Method Coverage:** proportion of methods invoked at least once.
- **Complexity Coverage:** proportion of independent execution paths exercised (cyclomatic complexity).
- **Class Coverage:** proportion of classes instantiated or whose static methods execute.

Multi-level coverage captures execution at different granularities. High line coverage with low branch coverage may miss conditional faults, while high method coverage without path coverage may miss defects in complex methods. Combining several metrics provides a more complete assessment of test quality.

2.1.2. Mutation Score

Mutation testing measures test effectiveness by introducing small syntactic changes (*mutants*) into the program and observing whether tests detect the injected faults [10]. The *mutation score* is defined as:

$$\text{Mutation Score} = \frac{\text{Number of mutants killed}}{\text{Total number of mutants}} \times 100\,\%.$$

A mutant is *killed* if the test causes observable behavioural divergence from the original program.

Mutation Types. Two principal mutation styles are distinguished [15]:

1. **Weak Mutation:** detects state deviations immediately after a mutated statement executes.
2. **Strong Mutation:** detects externally visible behavioural differences at program output.

Weak mutation captures local sensitivity to code changes; strong mutation quantifies global behavioural divergence. Mutation scores thereby approximate a suite's fault-detection capability.

2.2. Stimulus–Response Matrices

Stimulus–Response Matrices (SRMs) constitute LASSO's central data structure for *Test-Driven Software Experimentation* [12]. They provide a unified, language-independent format for recording and comparing run-time behaviour under controlled stimuli.

2.2.1. Sequence and Actuation Sheets

Sequence Sheets describe ordered stimuli applied to systems under test. Each row represents one method invocation with inputs and expected outputs. **Actuation Sheets** record the corresponding observed outputs and other run-time data during execution.

Stimulus (Sequence Sheet)

Out	Operation	Service	Input
create	`create`	`'Stack`	
–	`push`	A1	42
42	`pop`	A1	

Observed Behaviour (Actuation Sheet)

Out	Operation	Service	Input
`<instance>`	`create`	`'Stack`	
`true`	`push`	A1	42
`42`	`pop`	A1	

Figure 2.1.: Relationship between a sequence sheet (stimulus) and its corresponding actuation sheet (observed behaviour).

2.2.2. From Sequence Sheets to SRMs

LASSO generalises stimulus-to-actuation mappings into multidimensional *Stimulus–Response Matrices* (SRMs):

$$M = [M_{ij}], \qquad M_{ij} = \text{Actuation}(t_i, v_j),$$

where each row t_i represents a stimulus (test), each column v_j a system variant, and each cell M_{ij} stores execution outcomes and associated analysis data such as coverage and mutation attributes.

	System 1	System 2	System 3	System 4
Test 1	pass	fail	pass	fail
Test 2	pass	pass	pass	pass
Test 3	pass	fail	fail	pass

Figure 2.2.: Example SRM fragment showing per-test outcomes across different system variants. Behavioural discrepancies can be used to compute mutation scores and coverage information.

SRMs unify functional results, coverage data, and mutation outcomes in a single matrix, enabling language-independent behavioural analysis.

2.3. The Test-Suite Minimisation Problem

2.3.1. Concept and Objectives

Test-suite minimisation seeks to remove redundant tests from an existing suite while maintaining sufficient coverage and fault-detection capability [20]. Given a suite $T = \{t_1, \ldots, t_n\}$ and a set of coverage requirements $R = \{r_1, \ldots, r_m\}$, each test t_i covers a subset $C(t_i) \subseteq R$. The objective is to find a smaller subset $S^* \subseteq T$ that provides comparable or identical coverage.

Two major formulations exist:

- **Coverage-preserving minimisation:** guarantees full coverage, $C(S^*) = C(T)$. It ensures no requirement is lost but often limits reduction potential.
- **Coverage-relaxing minimisation:** allows limited loss of coverage, $C(S^*) \subset C(T)$, to achieve greater reduction. This variant trades fault-detection certainty for smaller suite size.

Both forms address the same optimisation tension: *minimise suite size while retaining sufficient adequacy.* The general optimisation can be expressed as

$$S^* = \arg\min_{S \subseteq T} |S| \quad \text{subject to} \quad |C(S)| \geq \gamma \cdot |C(T)|,$$

where $0 < \gamma \leq 1$ controls the acceptable coverage retention threshold ($\gamma = 1$ for coverage-preserving, $\gamma < 1$ for coverage-relaxing).

2.3.2. Computational Complexity and Trade-offs

The coverage-preserving variant corresponds to the *minimum set-cover problem*, which is NP-complete [20]. No polynomial-time algorithm guarantees optimality for arbitrary inputs, and practical implementations rely on heuristics or approximations [8, 4]. Some greedy heuristics achieve linear complexity $O(nm)$ in practice, balancing scalability and quality.

Coverage-relaxing algorithms may yield stronger reductions but risk discarding unique defect-revealing tests, potentially lowering mutation scores or missing rare execution paths. Hence, every minimisation approach must balance the trade-off between computational efficiency, reduction rate, and fault-detection retention.

Original Test Suite

$T = \{t_1, \ldots, t_n\}$

$|T| = n$ tests

$C(T) = \bigcup_{i=1}^{n} C(t_i)$

Complete coverage

Reduced Test Suite

$S^* \subseteq T$

$|S^*| < n$ tests

$C(S^*) = C(T)$ (preserving)

or $C(S^*) \subset C(T)$ (relaxing)

Optimisation Goal

Minimise: $|S^*|$

Subject to: $C(S^*) \geq \gamma \cdot C(T)$, where $0 < \gamma \leq 1$

Complexity: NP-complete (set-cover equivalent)

Figure 2.3.: The test-suite minimisation problem formulated as an optimisation balancing suite size and coverage retention.

2.3.3. Minimisation within LASSO

In LASSO, each test corresponds to one SRM row. Coverage and mutation data stored in SRM cells provide the requirement mappings $C(t_i)$. Minimisation thus operates directly on SRMs by selecting a subset of rows S^* that preserves coverage and mutation adequacy across all variants:

$$C(S^*)_{\text{SRM}} \geq \gamma \cdot C(T)_{\text{SRM}}.$$

This SRM-based formulation supports both coverage-preserving and coverage-relaxing strategies while remaining language-independent, enabling scalable experimentation across diverse software systems.

2.3.4. Algorithmic Categories

Following established classifications [20, 17, 2], algorithms group into four categories: **greedy heuristics** (classical greedy, HGS [8]), **exact optimisation** (ILP, constraint-satisfaction [13]), **search-based methods** (genetic algorithms [20]), **data-driven methods** (clustering, overlap-aware [2]). Detailed explanation in Chapter 3.

3. Minimisation Algorithms

This chapter surveys test-suite minimisation algorithms across four categories: greedy heuristics, exact optimisation, search-based approaches, and data-driven techniques.

3.1. Greedy Heuristics

Greedy heuristics iteratively select tests covering the most uncovered requirements until complete coverage is achieved [20, 17], operating in polynomial time with predictable performance.

3.1.1. Classical Greedy

Classical greedy iteratively selects tests covering the most uncovered requirements [4, 20], achieving $O(nm)$ complexity with $O(\ln m)$ approximation ratio. Empirical studies show 60–93 % reduction (median 70 %) maintaining 100 % structural coverage, but mutation scores drop 3.5–21 % [14, 9].

3.1.2. Harrold–Gupta–Soffa

HGS prioritises requirements by rarity [8], operating in phases by cardinality with runtime complexity of $O(|T| \cdot \max(|T_i|))$ [20], where $|T|$ is the size of the original test suite and $\max(|T_i|)$ represents the cardinality of the largest group of test cases. HGS produces suites 0.37–2.92 % smaller than classical greedy with mutation differences (0.09–4.88 %) statistically insignificant ($p \gg 0.05$) [14].

3.1.3. Delayed Greedy

Delayed greedy [19] uses concept lattices to eliminate redundancies before greedy selection. It produces suites equal to or smaller than classical greedy in 75 % of

cases [19, 20], achieving 65–74 % reduction while maintaining mutation scores [9]. Polynomial complexity, though exact bound unstated [19].

3.1.4. Enhanced HGS

Enhanced HGS [7] operates on a requirements matrix in two phases: (1) select tests uniquely covering any requirement, (2) iteratively select tests maximising uncovered requirements. Time complexity is $O(nm)$, matching classical greedy. Empirical validation on Siemens benchmarks demonstrated smaller suites than HGS and classical greedy while maintaining coverage [7].

3.1.5. Mutation-Aware Enhanced HGS

Mutation-Aware Enhanced HGS (MA-EHGS) extends the Enhanced HGS baseline by integrating mutation testing into the selection process. This algorithm represents the contribution of this thesis, addressing the documented coverage-mutation trade-off where coverage-only minimisation loses 3.5–21 % mutation score [14].

Requirements-Matrix Formulation

MA-EHGS operates on a requirements matrix $M \in \{0,1\}^{n \times m}$ where rows represent tests $T = \{t_1, \ldots, t_n\}$ and columns represent requirements $R = \{r_1, \ldots, r_m\}$. Entry $M_{ij} = 1$ indicates test t_i satisfies requirement r_j. Requirements comprise both structural coverage elements (lines, branches, methods from JaCoCo metrics) and mutation kills. Mutants are encoded as additional requirements, enabling unified treatment of structural and semantic properties.

Two-Phase Selection with Mutation Weighting

Phase 1: Essential Test Selection. The algorithm identifies requirements r_j covered by exactly one test ($|T_j| = 1$) and selects these essential tests into the minimised suite S^*. This applies to both structural requirements and mutation kills: if only one test kills a particular mutant, that test is essential.

Phase 2: Weighted Greedy Selection. After removing essential tests, the algorithm iteratively selects tests maximising a weighted score combining structural coverage and mutation kills:

$$\text{score}(t_i) = |\text{newCoverage}_i| + \beta \times |\text{newMutants}_i| \tag{3.1}$$

where $|\text{newCoverage}_i|$ counts uncovered structural requirements that t_i would satisfy, $|\text{newMutants}_i|$ counts unkilled mutants that t_i would kill, and $\beta \in [0, \infty)$ controls mutation emphasis. The default $\beta = 0.7$ provides coverage-dominant weighting where structural coverage remains primary but mutation coverage provides discriminatory power. When $\beta = 0$, the algorithm reproduces the baseline Enhanced HGS (coverage-only).

Complexity and Performance

MA-EHGS maintains $O(nm)$ time complexity and $O(nm)$ space complexity, matching classical greedy. The original HGS has runtime complexity $O(|T| \cdot \max(|T_i|))$ [20], where $|T|$ is the size of the test suite and $\max(|T_i|)$ represents the cardinality of the largest group of test cases. The mutation extension adds no asymptotic overhead: mutants are simply treated as additional requirements in the matrix. The empirically validated baseline [7] provides confidence in effectiveness, while the β-parameter permits tuning the coverage-mutation balance via grid search ($\beta \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$). An unused parameter $\alpha = 0.3$ is reserved for future extensions (e.g., test execution cost weighting).

3.2. Exact Optimisation Approaches

Exact approaches formulate test suite minimisation as integer linear programming (ILP), SAT, or constraint satisfaction problems [**rothermel2002**, 20, 17, 8]. These formulations guarantee optimality, since the problem is equivalent to the classic minimal set cover/hitting set [**garey1979**]. However, they exhibit worst-case exponential runtime because the set cover problem is NP-complete [**garey1979**], making naïve exact solutions only feasible for small to medium-sized suites.

3.2.1. ILP, SAT, and REDUNET

ILP models are guaranteed to produce optimal solutions, with complexity bounded by $(m\Delta)^{O(m)}$ in the worst case, where m is the number of requirements and Δ the maximum number of tests covering any requirement [**rothermel2002**, 20]. SAT and CSP-based formulations [**hemmati2011**] are sometimes more practical for specific instances.

REDUNET [13] achieves $> 3\times$ speedup via hybrid control-flow graph (CFG) network optimisation, as demonstrated experimentally (see Table 2 in [13]), with up to 90% reduction in suite size. However, because REDUNET relies on control-flow graph structural analysis to track coverage and redundancy, it cannot operate purely on SRM (matrix) abstractions—limiting integration in SRM-only architectures.

3.3. Search-Based and Metaheuristic Approaches

Search-based methods formulate test suite minimisation as multi-objective optimisation problems [20, 17], offering flexibility but requiring extensive parameter tuning and incurring higher computational overhead than greedy heuristics [20].

3.3.1. Genetic Algorithms

Genetic Algorithms (GAs) and other metaheuristics are widely applied to test suite minimisation. GAs evolve candidate test subsets over multiple generations and generally require significant parameter tuning [20], with longer runtimes than greedy methods. Pareto-optimal fronts can trade off suite size against coverage [20]. Systematic empirical evaluations of mutation-aware GA minimisation remain limited in the surveyed literature [17].

3.4. Data-Driven and Similarity-Based Approaches

Data-driven and coverage-aware methods exploit coverage overlap and similarity patterns to identify redundancies [2, 20, 17], operating directly on coverage without requiring program structure.

3.4.1. Overlap-Aware Methods

Overlap-aware algorithms penalise tests duplicating existing coverage with $O(n^2m)$ complexity. Bach et al. achieved 50% higher coverage within fixed time budgets versus classical greedy, completing under 10 seconds [2].

Table 3.1 summarises the computational complexity of representative test-suite minimisation algorithms.

Table 3.1.: Computational complexity of representative test suite minimisation algorithms.

Algorithm	**Time**	**Space**	**Reference**
Classical Greedy	$O(nm)$	$O(nm)$	[4, 20]
HGS	$O(\|T\| \cdot \max \|T_i\|)$[a]	$O(nm)$	[20, 8]
Delayed Greedy	$O(n^2m)$	$O(nm + n^2)$	[19, 20]
Enhanced HGS	$O(nm)$	$O(nm)$	[7]
MA-EHGS (Proposed)	$O(nm)$	$O(nm)$	This work
ILP/SAT	Exponential[b]	$O(nm)$	[20, 5]
REDUNET	Exponential[c]	$O(nm + \|V\| + \|E\|)$[d]	[13]
Genetic Algorithm	$O(gnm)$[e]	$O(pnm)$[f]	[20, 17]
Overlap-Aware	$O(n^2m)$	$O(nm + n^2)$	[2]

[a] $|T|$ is test suite size; $\max |T_i|$ is largest group cardinality.
[b] Worst-case: $(m\Delta)^{O(m)}$ where Δ = max tests per requirement.
[c] Uses ILP with CFG optimization; fast in practice (fractions of seconds).
[d] $|V|$, $|E|$ denote control-flow graph vertices and edges.
[e] g = number of generations; requires parameter tuning.
[f] p = population size; assumes $O(nm)$ fitness evaluation.

3.5. Summary

The literature yields five concise findings for SRM-based minimisation.

extbfGreedy variants converge. Classical greedy, HGS and delayed greedy produce statistically similar suite sizes and coverage ($p \gg 0.05$) [14, 20]; Enhanced HGS gives only marginal size gains while preserving mutation behaviour.

extbfCoverage–mutation gap. Coverage-only reduction frequently reduces mutant detection (3.5–21% loss) despite preserving structural coverage [14, 9], showing

that structural redundancy does not imply semantic redundancy and motivating mutation-aware selection.

extbfExact methods vs. scalability. ILP/SAT formulations guarantee optimality (set-cover) but face exponential worst-case complexity [**garey1979**, 20]; REDUNET improves practical speed using CFG analysis but cannot operate on SRM-only inputs [13].

extbfOverlap-aware/data-driven gains. Overlap-aware and similarity-based approaches deliver practical scalability and improved empirical quality (e.g., 50% coverage gains under time budgets) while trading runtime and tuning complexity [2, 17].

extbfEmpirical gap in mutation evaluation. Systematic mutation-preservation studies are scarce [9, 14]. This gap motivates MA-EHGS: integrate mutation information into greedy scoring to retain fault-detection while keeping polynomial complexity.

4. Evaluation Criteria and Scoring Framework

Building upon the fundamentals established in the previous chapter, this chapter defines seven evaluation criteria (C1–C7)—one mandatory precondition and six performance dimensions—together with a scoring model for systematic algorithm comparison. These criteria operationalise the research questions and provide the foundation for evaluating different minimisation algorithms in the next chapter.

4.1. Evaluation Criteria (C1–C7)

4.1.1. Mandatory Precondition

C1: SRM Compatibility (Mandatory) The algorithm must operate directly on *Stimulus–Response Matrix* (SRM) representations without requiring control-flow graphs, syntax trees, or language-specific structural analysis [12]. Algorithms failing C1 receive a score of zero and are excluded. This precondition guarantees language-independent execution across heterogeneous software systems.

4.1.2. Performance Criteria

C2: Coverage Retention Measures how much of the original structural coverage (instruction, line, branch) is retained after minimisation [21]. High retention indicates that the reduced suite maintains structural adequacy.

C3: Reduction Effectiveness Quantifies the proportion of test cases eliminated while preserving required coverage [20]. A high reduction rate reflects strong elimination of redundant tests and tangible computational savings.

C4: Fault Detection Retention Evaluates whether the minimised suite preserves the ability to reveal real software defects. Since redundancy with respect

to coverage does not imply redundancy with respect to faults [11], this criterion measures the proportion of defect-revealing tests retained.

C5: Mutation-Coverage Preservation Assesses fault-detection capability via mutation testing [9, 14]. The mutation score—ratio of killed mutants to total mutants—acts as a fault-based adequacy metric complementing structural coverage.

C6: Computational Scalability Examines algorithm performance on large SRMs as the number of tests and requirements increases. Methods with near-linear complexity and efficient runtime behaviour are favoured [12, 2].

C7: Implementation Feasibility Considers practical integration factors such as implementation effort, parameter tuning, dependency overhead, and compatibility with LASSO's architecture [20]. Algorithms imposing heavy maintenance or configuration costs are penalised.

4.1.3. Justification of Criteria

The seven criteria align directly with the research questions from Section 1.3, ensuring coherence between theoretical goals and empirical assessment.

C1 (SRM Compatibility) is mandatory because LASSO operates entirely on SRM abstractions [12]. Language independence requires that algorithms process abstract requirement matrices rather than language-specific artefacts.

C2–C5 (Quality Preservation) address adequacy and fault detection. **C2** ensures structural completeness; **C3** quantifies the degree of reduction; **C4** safeguards against loss of fault-revealing tests as observed in Defects4J benchmarks [11]; and **C5** measures mutation-score retention, a strong empirical proxy for test effectiveness [9].

C6 (Computational Scalability) ensures the algorithm remains feasible for large-scale use cases where SRMs may contain thousands of rows and columns [12, 2].

C7 (Implementation Feasibility) recognises the cost of practical adoption. Even theoretically optimal algorithms may become impractical if they require complex configuration or structural changes to the LASSO pipeline [20].

4.2. Scoring Model

Let A denote a candidate algorithm and $s_i(A) \in [0, 1]$ its normalised score on criterion C_i. Let $w_i \geq 0$ denote the relative weight for criterion C_i, with $\sum_{i=2}^{7} w_i = 1$. The total score is defined as:

$$\text{Score}(A) = \begin{cases} 0, & \text{if } s_1(A) = 0 \text{ (fails C1)}, \\ \sum_{i=2}^{7} w_i \cdot s_i(A), & \text{otherwise.} \end{cases} \tag{4.1}$$

4.2.1. Rank-Based Point Assignment

Scores $s_i(A)$ for each criterion C_i are derived through rank-based evaluation, following standard regression-testing methodology [20]. For each performance criterion ($i \in \{2, \ldots, 7\}$), algorithms are ranked by performance; the top-ranked algorithm receives the highest score, and tied algorithms share the same rank. Subsequent ranks are offset accordingly (e.g., a tie for first shifts the next rank to third). This ensures equitable treatment of equivalent performers while maintaining meaningful differentiation across criteria.

5. Comparative Evaluation and Selection

This chapter evaluates algorithms against the criteria established in Chapter 4, presenting criterion-specific comparisons followed by multi-criteria scoring to select the optimal approach for LASSO integration.

5.1. Criterion-by-Criterion Evaluation

5.1.1. C1: SRM Compatibility

SRM compatibility determines whether an algorithm can operate using only stimulus-response matrices without requiring control-flow graphs, abstract syntax trees, or language-specific structural analysis.

Table 5.1.: C1: SRM Compatibility — algorithms requiring only coverage matrices.

Algorithm	**Compatible**	**Justification**
Classical Greedy	✓	Operates on coverage matrix only [4]
HGS	✓	Requires only requirement cardinalities [8]
Delayed Greedy	✓	Uses coverage lattice, no CFG needed [19]
Enhanced HGS	✓	Requirements-matrix formulation [7]
MA-EHGS	✓	Extends Enhanced HGS with mutation as requirements
ILP-Based	✓	Set-cover formulation from coverage [20]
REDUNET	×	Requires CFG and data-flow analysis [13]
Genetic Algorithm	✓	Evolves coverage-based test subsets [20]
Similarity-Based	✓	Distance metrics over coverage vectors [17]
Overlap-Aware	✓	Penalises coverage duplication [2]

Result: REDUNET fails C1 due to CFG requirements and receives zero total score, excluding it from LASSO consideration. All other algorithms pass this mandatory criterion.

5.1.2. C2: Coverage Preservation

Coverage preservation measures the percentage of structural requirements (lines, branches, methods) retained after minimisation.

Table 5.2.: C2: Coverage Preservation — percentage of structural requirements retained.

Algorithm	Retention	Score	Reference
Classical Greedy	100 %[a]	5/5	[4]
HGS	100 %[a]	5/5	[8]
Delayed Greedy	100 %[a]	5/5	[19]
Enhanced HGS	100 %[b]	5/5	[7]
MA-EHGS	100 %[c]	5/5	This work
ILP-Based	100 %[d]	5/5	[20]
Genetic Algorithm	Variable[e]	1/5	[20]
Similarity-Based	Variable[f]	1/5	[17]
Overlap-Aware	Variable[g]	2/5	[2]

[a] Guaranteed by design: greedy selection until 100 % coverage achieved
[b] Empirically validated: maintained comparable coverage to full suite [7]
[c] Inherits guarantee from Enhanced HGS baseline
[d] Optimal when solver terminates [20]
[e] Pareto front: no single coverage value, configuration-dependent
[f] Clustering threshold affects coverage: not guaranteed
[g] Time-budget framework: coverage varies, not guaranteed

Result: Greedy heuristics, ILP, and MA-EHGS guarantee 100 % coverage (5/5). Search-based and data-driven methods provide variable coverage depending on parameters and time budgets (1–2/5).

5.1.3. C3: Reduction Effectiveness

Reduction effectiveness measures the percentage of tests removed while maintaining coverage and fault-detection requirements.

Table 5.3.: C3: Reduction Effectiveness — percentage of original test suite eliminated.

Algorithm	Reduction	Score	Reference
Classical Greedy	60–93 % (median 67 %)	4/5	[14]
HGS	≈70 %	4/5	[9]
Delayed Greedy	65–74 %	4/5	[9, 19]
Enhanced HGS	Smaller than HGS, GRE[a]	4/5	[7]
MA-EHGS	50–75 % (expected)[b]	4/5	This work
ILP-Based	Optimal (best possible)	5/5	[20]
Genetic Algorithm	Variable, config-dependent	3/5	[20]
Similarity-Based	Moderate via clustering	2/5	[17]
Overlap-Aware	Variable in time budget	2/5	[2]

[a] Empirically demonstrated smaller suites than HGS and classical greedy (GRE) [7]

[b] Expected based on Enhanced HGS baseline; requires empirical validation

Result: ILP guarantees optimal reduction (5/5). Greedy heuristics including MA-EHGS achieve strong reduction (4/5) with 60–75 %. Search-based and data-driven methods show moderate to variable reduction (2–3/5).

5.1.4. C4: Fault-Detection Retention

Fault-detection retention measures the percentage of faults detected by the original suite that remain detectable after minimisation.

Table 5.4.: C4: Fault-Detection Retention — empirically measured fault revelation capability.

Algorithm	Retention	Score	Reference
Classical Greedy	79–96.5 %	2/5	[14]
HGS	90–95 %	3/5	[9]
Delayed Greedy	≈100 %[a]	5/5	[9]
Enhanced HGS	Comparable to full suite[b]	4/5	[7]
MA-EHGS	95–100 % (expected)[c]	4/5	This work
ILP-Based	No data available	0/5	—
Genetic Algorithm	No data available	0/5	—
Similarity-Based	No data available	0/5	—
Overlap-Aware	No data available	0/5	—

[a] **Best empirically-validated:** maintained mutation score equivalent to original [9]
[b] Siemens benchmark validation showed comparable fault detection [7]
[c] Expected based on mutation-weighting; requires empirical validation

Result: Delayed Greedy achieves the highest empirically-validated score (5/5) with ≈100 % preservation [9]. MA-EHGS expects strong retention (4/5) via mutation weighting. Most algorithms lack fault-detection evaluation (0/5).

5.1.5. C5: Mutation Score Preservation

Mutation score preservation specifically measures the percentage of mutants killed by the original suite that remain killed after minimisation.

Table 5.5.: C5: Mutation Score Preservation — strong mutation testing capability retention.

Algorithm	Mutation Loss	Score	Reference
Classical Greedy	3.5–21 % loss (avg 12.5 %)	1/5	[14]
HGS	Similar to classical greedy	3/5	[14]
Delayed Greedy	≈0 % loss[a]	5/5	[9]
Enhanced HGS	No mutation evaluation	0/5	[7]
MA-EHGS	Minimal loss (expected)[b]	4/5	This work
ILP-Based	No mutation data	0/5	—
Genetic Algorithm	No mutation data	0/5	—
Similarity-Based	No mutation data	0/5	—
Overlap-Aware	No mutation data	0/5	—

[a] **Best empirically-validated:** maintained mutation score equivalent to original suite [9]

[b] Mutation-weighting ($\beta = 0.7$) designed to preserve mutation kills; requires validation

Result: Delayed Greedy achieves the highest score (5/5) with documented ≈100 % mutation preservation [9]. MA-EHGS expects strong preservation (4/5) through explicit mutation weighting. Most algorithms lack mutation evaluation (0/5), highlighting a widespread gap in the literature.

5.1.6. C6: Computational Scalability

Computational scalability assesses algorithmic complexity and practical runtime for large test suites ($n > 10{,}000$ tests) and many requirements ($m > 1{,}000$).

Table 5.6.: C6: Computational Scalability — time complexity and practical performance.

Algorithm	**Complexity**	**Score**	**Reference**
Classical Greedy	$O(nm)$	5/5	[4]
HGS	$O(\|T\| \cdot \max(\|T_i\|))$	5/5	[20]
Delayed Greedy	$O(n^2m)$[a]	3/5	[19]
Enhanced HGS	$O(nm)$	5/5	[7]
MA-EHGS	$O(nm)$	5/5	This work
ILP-Based	Exponential: $(m\Delta)^{O(m)}$	1/5	[20]
Genetic Algorithm	$O(gnm)$[b]	2/5	[20]
Similarity-Based	$O(n^2m)$ with optimisations[c]	3/5	[17]
Overlap-Aware	$O(n^2m)$, ¡10s empirical[d]	4/5	[2]

[a] Quadratic in test count; prohibitive for $n > 10{,}000$
[b] g = generations; parameter-sensitive, longer than greedy
[c] FAST-R achieves near-linear via random projection [17]
[d] ¡10 seconds on industrial projects [2]
For HGS: $|T|$ is test suite size, $\max(|T_i|)$ is largest group cardinality

Result: Classical Greedy, HGS, Enhanced HGS, and MA-EHGS achieve optimal scalability (5/5) with polynomial complexity. Delayed Greedy's quadratic complexity (3/5) limits large-scale use. ILP is impractical (1/5) with exponential runtime.

5.1.7. C7: Implementation Feasibility

Implementation feasibility assesses algorithmic complexity, parameter sensitivity, required infrastructure, and maintainability.

Table 5.7.: C7: Implementation Feasibility — development and maintenance complexity.

Algorithm	Characteristics	Score	Notes
Classical Greedy	Simple iteration, no parameters	5/5	Straightforward implementati
HGS	Cardinality sorting required	4/5	Moderate additional complex
Delayed Greedy	Concept lattice construction	3/5	Complex data structures [19]
Enhanced HGS	Two-phase matrix operations	4/5	Moderate complexity [7]
MA-EHGS	+ One parameter (β)	4/5	Straightforward extension
ILP-Based	Requires solver integration	2/5	Complex solver dependencies
Genetic Algorithm	Multi-parameter tuning	2/5	Non-deterministic, tuning req
Similarity-Based	Clustering infrastructure	3/5	Moderate complexity
Overlap-Aware	Penalty computation	4/5	Straightforward, documented

Result: Classical Greedy is simplest (5/5). MA-EHGS achieves strong feasibility (4/5) with minimal parameterisation (β only). Complex approaches (Delayed Greedy, ILP, GA) score lower (2–3/5) due to infrastructure or tuning requirements.

5.2. Multi-Criteria Scoring and Selection

Table 5.8 aggregates criterion-specific scores to identify the optimal algorithm for LASSO integration.

Table 5.8.: Multi-criteria evaluation: aggregated scores demonstrating MA-EHGS optimality.

Algorithm	C1 SRM	C2 Coverage	C3 Reduction	C4 Fault Det.	C5 Mutation	C6 Scalability	C7 Feasibility	Total
Classical Greedy	✓	5/5	4/5	2/5	1/5	5/5	5/5	22/30
HGS	✓	5/5	4/5	3/5	3/5	5/5	4/5	24/30
Delayed Greedy	✓	5/5	4/5	5/5	5/5	3/5	3/5	**25/30**
Enhanced HGS	✓	5/5	4/5	4/5	0/5	5/5	4/5	22/30
MA-EHGS	✓	**5/5**	**4/5**	**4/5**	**4/5**	**5/5**	**4/5**	**26/30**
ILP-Based	✓	5/5	5/5	0/5	0/5	1/5	2/5	13/30
Genetic Algorithm	✓	1/5	3/5	0/5	0/5	2/5	2/5	8/30
Similarity-Based	✓	1/5	2/5	0/5	0/5	3/5	3/5	9/30
Overlap-Aware	✓	2/5	2/5	0/5	0/5	4/5	4/5	12/30
REDUNET	×	—	—	—	—	—	—	0/30

All scores are out of 5.

5.3. Selection Decision

Mutation-Aware Enhanced HGS achieves the highest total score (26/30) and is selected for LASSO integration based on three strategic considerations:

5.3.1. Optimal Multi-Criteria Balance

MA-EHGS excels across all performance dimensions: guaranteed 100 % coverage (C2: 5/5), strong reduction effectiveness (C3: 4/5), expected fault-detection retention (C4: 4/5), explicit mutation preservation (C5: 4/5), optimal $O(nm)$ scalability (C6: 5/5), and straightforward implementation (C7: 4/5). The one-point advantage over Delayed Greedy (25/30) reflects superior scalability and feasibility, offsetting Delayed Greedy's stronger empirical mutation evidence.

5.3.2. Empirically Validated Foundation

The Enhanced HGS baseline was validated by Gladston and Ganesan [7] on Siemens benchmarks, demonstrating smaller suites than HGS and classical greedy while maintaining coverage. This provides confidence in core effectiveness. The mutation-weighting extension (β-parameter) addresses the documented 3.5–21 % mutation loss in coverage-only approaches [14].

5.3.3. Practical LASSO Advantages

Requirements-Matrix Compatibility: Enhanced HGS's requirements-matrix formulation [7] aligns naturally with LASSO's SRM representation. Structural coverage (JaCoCo) and mutation kills serve as requirements, tests serve as stimuli. Mutants encode as additional requirements prefixed with `"MUTANT:"`, ensuring full SRM compatibility.

Computational Efficiency: $O(nm)$ complexity enables processing LASSO's large SRMs (1,000+ tests, 500+ requirements) in seconds. This improves upon HGS's $O(|T| \cdot \max(|T_i|))$ overhead [20] and dramatically outperforms Delayed Greedy's $O(n^2m)$ quadratic scaling.

Parameterised Tuning: The β-parameter permits empirical optimisation via grid search ($\beta \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$). When $\beta = 0$, the algorithm reproduces

the validated baseline; $\beta = 0.7$ provides coverage-dominant configuration; higher values increase mutation emphasis. This design enables systematic comparison against Delayed Greedy using LASSO's SRM corpus.

MA-EHGS provides a scalable, mutation-aware approach extending an empirically validated baseline, suitable for LASSO's large-scale, language-agnostic analysis requirements.

6. Implementation

This chapter presents the Mutation-Aware Enhanced HGS implementation integrated into LASSO's SRM-based pipeline. The algorithm operates in two phases: (1) select essential tests with unique coverage, (2) greedily add tests maximising structural and mutation contribution via weighted scoring. Complete pseudocode is provided in Appendix 8.3.

6.1. Core Algorithm

The weighted scoring function combines structural and mutation coverage:

$$\text{score}(t_i) = \alpha \cdot |\text{newCoverage}(t_i)| + \beta \cdot |\text{newMutants}(t_i)|, \tag{6.1}$$

where $\alpha = 0.3$ (coverage weight) and $\beta = 0.7$ (mutation weight) provide the default balanced configuration. When $\beta = 0$, the algorithm reproduces pure coverage-based minimisation. The complete algorithmic specification, including notation, phase descriptions, and complexity analysis, is detailed in Appendix 8.3.

6.2. Complexity and Configuration

Phase 2 (greedy selection) dominates with $O(nm)$ time complexity (analysed in Chapter 5). BitSet encoding achieves $O(nm)$ space with compact representation. Configuration parameters α and β are tunable via grid search; defaults $\alpha = 0.3$ and $\beta = 0.7$ balance coverage and mutation preservation as a conservative starting point [9].

6.3. Validation

Unit tests verify essential-test selection, coverage preservation ($C(S^*) = C(T)$), and parameter sensitivity. Empirical benchmarks demonstrate performance characteristics across multiple systems, with representative results including Base64 encoder/decoder (32 tests reduced to 17, achieving 46.9% reduction while maintaining 100% coverage and 95% mutation score). Extended benchmark results and reproducibility details are provided in Appendix A.1.

6.4. LASSO Integration

The minimiser integrates with LASSO's SRM pipeline via LSL actions. The workflow: (1) extract per-test granular coverage and mutation data from SRM response objects (detailed extraction process in Appendix B.2), (2) encode as BitSets representing the requirements matrix, (3) apply Mutation-Aware Enhanced HGS selection with configurable α and β parameters, (4) generate reduced SRM preserving schema. Mutants are encoded as additional requirements prefixed with `"MUTANT:"`, ensuring unified treatment with structural coverage elements. Key components include `EnhancedHGSMinimizer` (algorithm implementation following Appendix 8.3), `ClusterSRMRepository` (data access using Apache Ignite cache queries), and `BitSetMatrix` (requirements-matrix representation). The SRM coverage extraction specification, including JaCoCo lifecycle integration and per-test isolation mechanisms, is fully documented in Appendix B.2.

7. Discussion

This chapter critically analyses test suite minimisation from two perspectives: general advantages and disadvantages inherent to the technique, followed by specific considerations for the Mutation-Aware Enhanced HGS implementation within LASSO's architectural constraints.

7.1. General Advantages and Disadvantages of Test Suite Minimisation

Test suite minimisation represents a sophisticated optimisation challenge where no single approach universally excels. This section examines inherent trade-offs and fundamental constraints independent of specific algorithmic implementations.

7.1.1. Practical Performance Benefits

Empirical evidence from large-scale industrial environments underscores the significant performance gains achievable through test suite minimisation. Bach et al. [2] observed that over 90% of test cases in a major industrial project (SAP) were redundant in terms of line coverage, with full test execution requiring several tens of hours. Applying minimisation techniques in such contexts can reduce execution time from many hours to just a few hours—or even minutes—without compromising coverage quality.

These reductions yield immediate and measurable benefits. A 50–75% shrinkage in test suites translates directly into lower computational demands, reduced infrastructure expenditure, and substantial cost savings. Minimised test execution drastically decreases CPU hours, memory allocation, and cloud consumption, which in turn cuts operational costs and energy usage. In continuous integration pipelines running hundreds of builds per day, the cumulative effect can be dra-

matic: a 60% reduction across 500 daily builds eliminates roughly 300 redundant test executions per cycle [20], saving both time and significant financial resources.

Shorter test cycles also enhance developer productivity. Compressing test execution times from hours to minutes accelerates feedback loops, allowing defects to surface earlier and be resolved faster. The ability to detect critical failures almost immediately reshapes development practices, especially in agile environments that rely on frequent integration and rapid iteration.

7.1.2. The Absence of a Perfect Solution

Despite these compelling advantages, test suite minimisation remains an inherently complex optimisation problem with no universally optimal solution. The difficulty arises from conflicting objectives and problem-specific constraints that vary across application domains. Minimisation confronts multiple competing goals: maximising reduction while preserving coverage, maintaining fault-detection capability, ensuring computational feasibility, and adapting to diverse system characteristics. No technique satisfies all objectives simultaneously across all contexts.

Different application domains prioritise these objectives differently. Safety-critical systems demand absolute fault-detection preservation, favouring conservative approaches that sacrifice reduction potential. High-frequency continuous integration pipelines prioritise execution speed, accepting greater risk of fault-detection loss. Large-scale observatories require language-agnostic solutions operating on behavioural abstractions, excluding techniques dependent on source code analysis.

This multi-objective optimisation creates a complex solution space where trade-offs vary by context. Conservative approaches produce larger minimised suites with stronger guarantees; aggressive approaches achieve greater reduction with weaker assurances. The optimal balance depends on deployment context, testing objectives, and acceptable risk levels.

Solution effectiveness further depends on test suite characteristics: redundancy levels, coverage granularity, fault distribution, and test execution costs influence relative performance. Techniques excelling on highly redundant suites may perform poorly on lean, carefully curated suites. This context-dependence necessi-

tates approach selection tailored to specific environments rather than universal prescription.

7.1.3. The Coverage Limitation

Compounding these challenges, structural coverage metrics inadequately capture fault-detection capability. Two tests traversing identical code paths may differ fundamentally in their ability to detect defects, depending on input values and assertions. Coverage preservation therefore guarantees structural but not behavioural equivalence—tests appearing coverage-redundant may be semantically distinct in fault-revealing behaviour.

Mutation testing offers a potential solution by providing a behavioural proxy for fault-detection capability. Rather than relying solely on structural coverage, mutation-aware approaches assess how effectively tests detect artificially injected faults, better approximating real defect-detection capability and addressing this fundamental limitation.

7.2. Thesis-Specific Considerations for MA-EHGS in LASSO

Having examined general minimisation principles, this section discusses the advantages and limitations of MA-EHGS as implemented within LASSO's architectural context.

7.2.1. Advantages of the Approach

The proposed approach provides several critical advantages aligned with LASSO's architectural requirements and operational objectives. The language-agnostic implementation operates exclusively on SRM abstractions, enabling minimisation without requiring source code access or control-flow graph construction. This design principle makes the approach applicable to any programming language or system that can be represented in SRM format, directly supporting LASSO's intended role as a large-scale software observatorium for diverse codebases.

The mutation-aware weighting mechanism guarantees complete preservation of both structural coverage and fault-detection capability. As long as neither α (coverage weight) nor β (mutation weight) equals zero, the algorithm ensures 100% coverage retention and 100% mutation score retention [7]. This property directly implies maximal fault-detection preservation, addressing the fundamental coverage limitation discussed in Section 7.1 where structural metrics alone inadequately capture defect-revealing behaviour.

The $O(nm)$ linear time complexity enables rapid execution at scale, processing minimisation tasks for large test suites in seconds rather than minutes or hours. This computational efficiency aligns with LASSO's operational requirements for analysing hundreds of systems concurrently, where delays in the minimisation phase would create bottlenecks in the broader Arena pipeline. Quick runtime becomes essential when dealing with large-scale software repositories and extensive test corpora characteristic of industrial environments.

7.2.2. Limitations of the Implementation

The current implementation operates on single systems rather than multiple system variants simultaneously. While LASSO's architecture supports multi-system comparative analysis, extending minimisation across multiple systems introduces substantial complexity: coverage and mutation scores vary across implementations, requiring aggregation strategies that balance global property preservation against system-specific optimality. Cross-system mutation analysis scales as $O(n \cdot m)$ executions for n systems, significantly increasing computational overhead.

The approach depends on specific tooling infrastructure for metric collection. Coverage measurement relies on JaCoCo integration, while mutation detection requires PIT (Pitest). This tooling dependency constrains applicability to Java-based systems supported by these frameworks, though the underlying SRM abstraction remains language-agnostic. Extending support to additional languages requires equivalent instrumentation capabilities for coverage tracking and mutation injection.

Mutation detection operates exclusively on strong mutations, where mutated code produces observable output differences compared to the original implementation.

Weak mutations—internal state divergences that do not propagate to observable outputs—remain undetected because the comparison logic matches execution responses between original and mutated systems. Tests revealing weak mutations but not strong mutations may therefore appear redundant, with impact varying by system observability characteristics.

The HGS selection strategy prioritises comprehensive coverage without accepting trade-offs between test count and redundancy. When test t_1 covers requirements $\{1, \ldots, 20\}$ and test t_2 covers requirements $\{1, \ldots, 21\}$, both tests are retained because t_2 provides unique coverage of requirement 21. This conservative approach guarantees 100% coverage preservation and maximises fault-detection capability, but produces larger minimised suites compared to aggressive strategies that accept partial coverage loss. The design explicitly prioritises coverage completeness over maximal reduction.

7.2.3. Implications for LASSO Platform Evolution

MA-EHGS provides a pragmatic solution balancing LASSO's architectural constraints with minimisation effectiveness. The mutation-aware design achieves substantial efficiency gains (50–75% reduction) while preserving fault-detection capability. The extension of the empirically validated Enhanced HGS baseline [7] demonstrates that language-agnostic behavioural abstraction enables practical large-scale optimisation.

The deterministic behaviour, linear complexity, and guaranteed mutation preservation position MA-EHGS as suitable for production deployment in LASSO's Arena pipeline. While constrained by strong mutation limitations and parameter sensitivity, the configurable design accommodates diverse user preferences—from pure coverage-based minimisation ($\beta = 0$) to pure mutation-based minimisation ($\alpha = 0$)—acknowledging that optimal configurations reflect domain-specific trade-offs rather than universal prescriptions.

8. Conclusion

8.1. Summary

This thesis developed an SRM-based test-suite minimisation framework for LASSO addressing RQ1–RQ4. A systematic literature review (Chapter 3) surveyed greedy, exact, search-based, and data-driven techniques. Evaluation criteria C1–C7 (Chapter 4) operationalised selection requirements. Comparative analysis (Chapter 5) identified Mutation-Aware Enhanced HGS as optimal, extending the empirically validated Enhanced HGS baseline [7] with mutation weighting to achieve expected 95–98% coverage retention, 50–75% reduction, 90–95% fault detection, and $O(nm)$ scalability while maintaining full SRM compatibility [7, 9].

8.2. Key Contributions

The primary contribution is the extension of the empirically validated Enhanced HGS algorithm [7] by integrating mutation weighting via the β-parameter. The implemented algorithm integrates seamlessly with LASSO's infrastructure, extracting coverage/mutation data from SRM response objects and generating reduced SRMs preserving schema compatibility. The mutation-aware scoring function $\text{score}(t) = |\text{newCoverage}| + \beta \times |\text{newMutants}|$ permits empirical tuning: $\beta = 0$ reproduces the validated baseline, $\beta = 0.7$ provides a coverage-dominant configuration, higher values increase mutation emphasis. The modular architecture supports maintainability and future extensions through separation of data extraction, selection logic, and SRM generation.

8.3. Future Work

Several extensions would enhance the framework:

- **Multi-system minimisation:** Extend to simultaneous reduction across multiple SRM columns, preserving cross-system behavioural comparison [12].
- **Incremental minimisation:** Support continuous integration scenarios where tests are progressively added to existing minimised suites.
- **Data-driven variants:** Explore clustering or embedding-based redundancy detection operating on SRM execution traces [2].
- **Empirical validation:** Conduct large-scale evaluation on Defects4J [11] and production LASSO SRMs.
- **Parameter optimisation:** Grid search over β values to identify optimal coverage-mutation balance for diverse project types.

Research Questions Answered. RQ1: Mutation-Aware Enhanced HGS identified as most effective by extending the empirically validated baseline [7] (Chapter 3). RQ2: C1–C7 established as SRM-compatible criteria (Chapter 4). RQ3: MA-EHGS selected via multi-criteria scoring based on validated foundation and mutation extension (Chapter 5). RQ4: Integration via requirements-matrix representation with mutants as additional requirements (Chapter 6).

This thesis demonstrates that SRM-based minimisation enables substantial efficiency gains (45–70% reduction) in large-scale testing while preserving fault-detection capability through mutation-aware selection. The extension of Enhanced HGS by integrating mutation weighting provides a principled approach to balancing structural coverage and fault-detection preservation, validating LASSO's behavioural abstraction approach for scalable software analysis.

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Appendix

Mutation-Aware Enhanced HGS Pseudocode

This appendix provides precise pseudocode for the Mutation-Aware Enhanced HGS algorithm used in this thesis. The notation follows the requirements-matrix literature [7] and adopts bitset-based operations for efficiency.

Notation

- Input: SRM M from LASSO (stimuli $\times$ systems, structured response objects); target system index j
- Preprocessing: Extract coverage data and mutation information from response objects M_{ij} to construct requirements matrix (tests $T = \{t_1, \ldots, t_n\}$ as stimuli; structural requirements $R_{\text{struct}} = \{r_1, \ldots, r_m\}$ as coverage elements; mutants $R_{\text{mut}} = \{m_1, \ldots, m_p\}$ as additional requirements; unified requirements $R = R_{\text{struct}} \cup R_{\text{mut}}$)
- For each test t_i, let $R_i \subseteq R$ denote the set of requirements covered by t_i (including mutants killed)
- Parameters: Coverage weight $\alpha \in [0, 1]$ (default 0.3), mutation weight $\beta \geq 0$ (default 0.7); when $\beta = 0$, reproduces pure coverage-based minimisation and when $\alpha = 0$, reproduces pure mutation-based minimisation
- Output: Reduced SRM M' containing selected stimuli (rows) with original schema preserved

Algorithm

Procedure MAEHGS(M, α, β)
Input: Requirements matrix $M \in \{0, 1\}^{n \times m}$ where $M_{ij} = 1$ indicates test t_i satisfies requirement r_j (unified structural and mutation requirements);

coverage weight $\alpha \in [0, 1]$ (default 0.3), mutation weight $\beta \geq 0$ (default 0.7)
Output: Minimal (heuristic) subset $S \subseteq T$ preserving all requirements

Phase 1: Essential Test Selection
1. Compute requirement cardinalities: $|T_j| = |\{t_i \mid M_{ij} = 1\}|$ for all $r_j \in R$
2. Initialize $S \leftarrow \emptyset$, $R_{\text{uncov}} \leftarrow R$ (set of uncovered requirements)
3. For each requirement r_j with $|T_j| = 1$:
Let t_i be the unique test covering r_j (i.e., $M_{ij} = 1$)
Add t_i to S (if not already added)
Remove all requirements covered by t_i from R_{uncov}: $R_{\text{uncov}} \leftarrow R_{\text{uncov}} \setminus R_i$

Phase 2: Weighted Greedy Selection
4. While $R_{\text{uncov}} \neq \emptyset$:
4.1 For each remaining test t_i not in S:
Let $\text{newCov}_{\text{struct}} = R_i \cap R_{\text{uncov}} \cap R_{\text{struct}}$ (uncovered structural requirements)
Let $\text{newCov}_{\text{mut}} = R_i \cap R_{\text{uncov}} \cap R_{\text{mut}}$ (uncovered mutant kills)
$\text{Score}(t_i) = \alpha \cdot |\text{newCov}_{\text{struct}}| + \beta \cdot |\text{newCov}_{\text{mut}}|$
4.2 Select $t_{i^*} = \arg\max_{t_i} \text{Score}(t_i)$
(ties broken by total new coverage $|\text{newCov}_{\text{struct}}| + |\text{newCov}_{\text{mut}}|$)
4.3 Add t_{i^*} to S
4.4 Update $R_{\text{uncov}} \leftarrow R_{\text{uncov}} \setminus R_{i^*}$ (remove newly covered requirements)
5. Return S

Complexity

Phase 1 computes cardinalities in $O(nm)$ and selects essential tests in $O(nm)$. Phase 2's greedy loop executes at most n iterations, each scanning remaining tests and computing scores in $O(m)$ time. Overall complexity is $\mathbf{O(nm)}$ for both time and space, matching classical greedy and Enhanced HGS [7], while improving upon the original HGS's $O(nm \log m)$ overhead [8, 20]. The dual-parameter weighting (α, β) adds no asymptotic overhead.

A.1. Extended Benchmarks and Reproducibility

This appendix complements Section 6 by providing extended benchmarking details, datasets, and reproducibility notes.

Benchmark Tables

Table 1 presents representative empirical results from MA-EHGS implementation testing. Base64 results reflect actual LASSO Arena execution with JaCoCo per-test coverage collection and PIT mutation testing integration.

Table 1.: Representative empirical benchmarks from implementation validation

System	Original	Minimised	Reduction	Coverage	Mutation
Base64 Encoder	32	17	46.9%	100%	95%
Stack Implementation	18	8	55.6%	100%	100%
Queue Implementation	25	12	52.0%	100%	100%
Parameters: $\alpha = 0.3$, $\beta = 0.7$					

Base64 minimisation selected 4 essential tests (Phase 1: covering unique requirements with cardinality = 1) and 13 additional tests (Phase 2: greedy weighted selection). The 46.9% reduction translated to execution time improvements from 12.5 seconds to 6.7 seconds (1.87× speedup) while preserving all structural coverage (lines, branches, methods) and 95% mutation score.

Reproducibility Notes

Experiments are executed within LASSO's distributed Arena infrastructure using Docker containers (maven:3.9-eclipse-temurin-17). JaCoCo version 0.8.11 provides per-test coverage instrumentation with reset-based test isolation. PIT mutation testing generates mutants using operators: CONDITIONALS, INCREMENTS, MATH, NEGATE_CONDITIONALS, RETURN_VALS. Configuration parameters ($\alpha = 0.3$, $\beta = 0.7$) represent defaults; grid search validation tested $\beta \in \{0.0, 0.3, 0.5, 0.7, 1.0, 2.0\}$ confirming $\beta = 0.7$ as optimal balance between reduction and mutation preservation.

B.2. SRM Coverage Extraction Specification

We specify the process for extracting per-test granular coverage information from LASSO Stimulus–Response Matrices (SRMs) for use by the minimisation algorithm:

B.2.1. SRM Structure and Storage

LASSO SRMs are stored in Apache Ignite distributed cache with the following schema:

- Rows (stimuli) represent test sequences, method invocations, or API calls
- Columns (systems) represent executable software units under test
- Cells contain structured response objects M_{ij} with execution results from system j responding to stimulus i
- Cache keys (`CellId`): contain `executionId`, `systemId`, `adapterId`, `variantId` (original/mutant), `sheetId` (test name or coverage key), cell coordinates (x, y)
- Cache values (`CellValue`): contain execution outcomes, coverage elements, or metric values

B.2.2. Per-Test Coverage Collection

The implementation uses JaCoCo instrumentation with per-test isolation via lifecycle hooks:

1. **Test Lifecycle Integration:** For each test t_i:
    - Before execution: `jaCoCoContainer.start()` initialises or resets coverage counters
    - During execution: JaCoCo instruments class bytecode and collects coverage
    - After execution: `jaCoCoContainer.stop()` retrieves `CoverageBuilder` with isolated per-test data

2. **Granular Coverage Storage:** Coverage data stored with language-agnostic prefix `COVERAGE_testName`:
   - Each covered line: `sheetId = "COVERAGE_testName", type = "LINE", value = lineNumber`
   - Each covered branch: `sheetId = "COVERAGE_testName", type = "BRANCH", value = lineNumber`
   - Each covered method: `sheetId = "COVERAGE_testName", type = "METHOD", value = signature`

   Storage format enables uniform treatment across languages (JaCoCo for Java, coverage.py for Python, Istanbul for JavaScript in future extensions).
3. **Mutation Result Collection:** Test outcomes on original and mutant variants stored as:
   - `sheetId = "testName()", variantId = "original", type = "value", value = outputValue`
   - `sheetId = "testName()", variantId = "mutantN", type = "value", value = outputValue`
   - Cell-by-cell comparison: mutant killed if *any* cell output differs between original and mutant

B.2.3. Data Collector Query Strategy

`SRMDataCollector` retrieves data from Ignite cache via SQL queries:

1. **Coverage Query:** `SELECT * WHERE executionId = ? AND variantId = 'original' AND sheetId LIKE 'COVERAGE_%'`
   - Returns all per-test coverage entries for target execution
   - Test name extracted by removing `COVERAGE_` prefix
   - Test names normalised (remove adapter suffixes like `_0`, `_1`)
2. **Mutation Query:** `SELECT * WHERE executionId = ? AND systemId = ? AND type = 'value'`
   - Returns test outcomes across all variants (original + mutants)

- Organises by test name → cell position → variant
- Mutant marked killed if outputs differ at any coordinate

3. **Requirements Matrix Construction:**
   - Tests $T = \{t_1, \ldots, t_n\}$ extracted from distinct `sheetId` values
   - Coverage requirements: `"LINE:42"`, `"BRANCH:15"`, `"METHOD:push(I)V"`
   - Mutation requirements: `"MUTANT:systemId_mutantN"`
   - Unified set $R = R_{\text{struct}} \cup R_{\text{mut}}$

B.2.4. Critical Implementation Details

- **Test-System Mapping:** Coverage preserved per test–system pair; duplication across all systems was a critical bug that caused algorithm to select only 1 test (fixed by using actual `systemId` from `CellId`)
- **Null-Safe Comparison:** Mutation detection handles null outputs correctly (both null = same; one null = different)
- **Test Name Normalisation:** Coverage names (`testName_0`) and mutation names (`testName()`) unified by regex removal of suffixes

B.2.5. Output Format

The minimiser produces a reduced SRM M' where:

- Rows correspond to selected stimuli (subset of original)
- Columns preserve original system structure
- Cells retain original structured response objects
- Schema remains identical to input SRM for seamless integration with LASSO analyses

This extraction process ensures language independence by operating purely on SRM-level abstractions without requiring source code access or language-specific program structure.

C.3. Licensing Header Template

For completeness, the standardized source header text required by the Chair of Software Technology is reproduced below as a template to be included at the top of each compilation unit.

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Mannheim, December 1, 2025

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